

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 1.2: Indices — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.2: Indices
Driving Question	DRIVING QUESTION: How can writing numbers as powers and logarithms help us make sense of — and calculate with — the enormous and tiny quantities that appear in Kenyan farming, finance, engineering, and science?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Noticing the Pattern: Index Notation and the Maize Doubling Puzzle	We observed that a maize farmer in Kitale starting with 1 seed produces a sequence — 1, 2, 4, 8, 16, 32, 64, 128, 256 — where every term is double the previous one, and that by Season 20 the count exceeds one million, making the number extremely long and difficult to write or compute with by hand.	We learned that each term in the doubling sequence can be written as a power of 2 using index notation (base ^{exponent}), that the index tells us how many times the base is used as a factor, and that this notation is far more compact and meaningful than writing out a long string of multiplications.	We explained that mathematicians use index notation because it compresses repeated multiplication into a short, readable form that preserves the structure of the calculation — the base shows what is being multiplied and the index shows how many times, making it easy to compare and work with very large numbers that arise in farming, science, finance, and engineering.
Lesson 2 Laws of Indices I: Product Law and Quotient Law	Learners expanded power expressions (e.g., $2^3 \times 2^4$) into full repeated-factor form, counted total factors, and recorded results in a structured table. They also cancelled common factors in quotient expressions and applied both patterns to a Kitale maize-seed context and an M-Pesa transaction problem.	The Product Law ($a^m \times a^n = a^{m+n}$) and Quotient Law ($a^m \div a^n = a^{m-n}$) hold for any common base. Multiplying or dividing powers of the same base reduces to adding or subtracting exponents — a shortcut that makes working with the maize-seed doubling numbers far more efficient than writing out all factors.	When we multiply powers of the same base, we are really counting all the repeated factors together — so the exponents add. When we divide, we cancel equal factors from top and bottom — so the exponents subtract. This is why index notation is powerful: operations on large numbers become simple arithmetic on small exponents, exactly what scientists, engineers, M-Pesa agents, and Kenyan farmers need when numbers grow very large or very small.
Lesson 3 Laws of Indices II — Power, Zero, and	A single E. coli bacterium in a Nairobi microbiology lab doubles every 20 minutes. After 30 doublings (10 hours), the count	We derived three new laws: (1) Power Law — $(a^m)^n = a^{mn}$, so $(2^3)^2 = 2^6$; (2) Zero Index Law — $a^0 = 1$ for any non-zero base,	The three laws are consistent extensions of the Product and Quotient Laws from Lessons 1–2. They work because the

Negative Index Laws	reaches $2^{30} \approx 10^9$. We also observed that the maize-doubling sequence can be extended backwards: Season 0 gives 2^0 , and 'before Season 0' gives 2^{-1} , 2^{-2} , 2^{-3} .	proven using the Quotient Law ($a^m \div a^n = 1 = a^0$); (3) Negative Index Law — $a^{-n} = 1/a^n$, proven by extending the Quotient Law past zero.	pattern of multiplying and dividing powers is unbroken — even when exponents reach zero or go negative. These laws allow scientists to rewrite 2^{30} as $(2^{10})^3$, making computation manageable. However, index laws alone cannot answer the question 'what is n if $2^n = 10^9$?' — that requires a new tool (logarithms) which we will meet in Lesson 4.
Lesson 4 Applying the Laws of Indices: Multi-Step Computations, Algebraic Simplification, and Index Equations	Learners applied all five laws of indices to multi-step numerical and algebraic expressions drawn from SGR engineering distances and Rift Valley geothermal energy data. They solved index equations by matching bases and presented annotated step-by-step solutions to peers at the board.	All five laws of indices work together in multi-step computations. Index notation is not merely a compact writing system — it is a calculation engine that allows engineers at SGR and scientists at Olkaria to work efficiently with enormous and tiny quantities. Solving index equations requires expressing both sides with the same base. The laws have a clear boundary: they cannot solve equations like $2^x = 500,000$ when x is not a whole number.	The Kitale maize doubling sequence produces numbers so large ($2^{20} > 1,000,000$) that only index notation makes them manageable. The five laws of indices provide the rules for combining, simplifying, and solving expressions involving these numbers. This is exactly why scientists, engineers, and bankers compress huge quantities into short power expressions — and it is why the farmer's notebook calculation problem from Lesson 1 demanded a smarter tool. The remaining gap — solving $2^x = 500,000$ — will be filled by logarithms in Lesson 5.
Lesson 5 Bridging to Logarithms: Index and Log Notation	When we asked which season produced 256 seeds, we found that we already needed the index — and for huge numbers like 1,048,576 finding the index quickly was very hard. We also saw that the Richter scale number for the 2016 Baringo earthquake (5.9) is itself an index written as part of a power of 10.	The logarithm of a number N to base a is the index x such that a to the x equals N . Index form and logarithm form are two ways of writing the same relationship. We can convert fluently between them. The Richter scale is a real-world logarithm scale used by Kenyan geoscientists every day.	Today we began to answer why scientists, engineers, and bankers compress large quantities into short expressions: the logarithm lets you work with the index (a small, manageable number) rather than the enormous result. This is why the Richter scale says 5.9 rather than writing out 10 to the power 5.9 as a full number. Logarithms are not a new idea — they are the index question asked backward.
Lesson 6 Reading Common Logarithms from Tables and Calculators	We read common logarithm values for Kenyan quantities — including a seed count of 256, a population density of 25000, and a pesticide concentration of 0.004 — from four-figure mathematical tables and verified each result with a calculator.	Every common logarithm has two parts: a characteristic (the whole number, determined by the power of 10) and a mantissa (the decimal part, read from the table using significant digits). Numbers less than 1 have a negative characteristic written using bar notation.	Because logarithms are indices, and indices keep track of how many times we multiply by the base, the characteristic tells us the order of magnitude of the number while the mantissa captures its precise significant-digit value — together they give a compact, readable code for any positive number, exactly what the Kitale farmer needs to compare harvests across seasons without writing out millions of digits.
Lesson 7	Learners used mathematical tables and	Logarithm laws for multiplication ($\log MN =$	The Maize Doubling Puzzle farmer no

Applying Logarithms: Multiplication, Division, Powers, and Roots	calculators to compute 3750×4200 , $98560 \div 35$, $(1.08)^{10}$ (KCB compound interest), and $\sqrt[3]{614125}$ (Kericho tea-farm yield per silo) using the three logarithm laws — then verified their answers digitally.	$\log M + \log N$), division ($\log M/N = \log M - \log N$), and powers/roots ($\log M^n = n \log M$) transform slow, error-prone multiplication and root-finding into fast addition and subtraction — because logarithms are exponents, and the logarithm laws mirror the index laws established in Lessons 2–3.	longer needs a notebook full of enormous numbers: by converting to logarithms, any multiplication or root calculation collapses into a single-line addition or multiplication — the same compressive power that scientists, engineers, and bankers rely on every day.
Lesson 8 Summative Assessment, Final Explanation, and Model Consolidation	We observed that the Kitale farmer's maize seed count doubles every season, producing the sequence 1, 2, 4, 8, ..., 256 over 8 seasons, and exceeding 1,000,000 by Season 20. We observed that writing 256 as 2^8 is more compact than the full number, and that multiplying $2^8 \times 2^4$ can be done by simply adding the exponents to get 2^{12} . We also observed that log tables allow us to turn multiplication of large numbers (like KES 4,680 $\times$ 327) into addition of logarithms, and that the answer can be recovered using the antilog.	We learned that any number can be expressed in index form using a base and an exponent. We learned the five laws of indices (Product, Quotient, Power of a Power, Zero Index, and Negative Index) and how to apply them to simplify and evaluate expressions. We learned that logarithm notation is simply another way of writing index notation — if $10^3 = 1000$, then $\log_{10}(1000) = 3$. We learned how to read common logarithms and antilogs from mathematical tables, and how to use these to perform multiplication, division, powers, and roots of numbers without a calculator.	Indices and logarithms exist because quantities in the real world — maize seed populations, compound interest in a Nairobi bank, electrical power in a Kenya Power grid, and microscopic organisms in a Nairobi laboratory — grow or shrink at rates that make ordinary arithmetic impractical. Writing numbers as powers of a base compresses enormous quantities into short, comparable expressions and transforms multiplication into addition (via the Product Law and its logarithmic equivalent). This is why scientists, engineers, and bankers use these tools: they make sense of and calculate with quantities that would otherwise overflow a notebook, a calculator screen, or a human working memory. The Maize Doubling Puzzle is solved — and the tools that solved it extend to every domain of Kenyan life where quantities grow beyond ordinary scale.

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